

3 Theoretical Development and Grammar

Code Example

The following code block was used as the basis for the theoretical demonstration:

```
1 print("This is an example!!!!");
2 int b = 20;
3 float c = (500 * 4) / 20 + 1 - 10;
4 print(c);
5 print("hello");
6 string answer_5 = c;
```

Grammar with Left Factoring

$$\begin{aligned} \langle \text{program} \rangle &\rightarrow \langle \text{statement_list} \rangle \\ \langle \text{statement_list} \rangle &\rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle \mid \varepsilon \\ \langle \text{statement} \rangle &\rightarrow \langle \text{declaration} \rangle ; \mid \langle \text{print_statement} \rangle ; \\ \langle \text{declaration} \rangle &\rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle \\ \langle \text{type} \rangle &\rightarrow \text{int} \mid \text{float} \mid \text{string} \\ \langle \text{print_statement} \rangle &\rightarrow \text{print} (\langle \text{value} \rangle) \\ \langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \mid \langle \text{literal} \rangle \\ \langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\ \langle \text{expression}' \rangle &\rightarrow + \langle \text{term} \rangle \langle \text{expression}' \rangle \mid - \langle \text{term} \rangle \langle \text{expression}' \rangle \mid \varepsilon \\ \langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\ \langle \text{term}' \rangle &\rightarrow * \langle \text{factor} \rangle \langle \text{term}' \rangle \mid / \langle \text{factor} \rangle \langle \text{term}' \rangle \mid \varepsilon \\ \langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \mid \langle \text{identifier} \rangle \mid (\langle \text{expression} \rangle) \\ \langle \text{literal} \rangle &\rightarrow " \langle \text{content} \rangle " \\ \langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \mid \varepsilon \\ \langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\ \langle \text{identifier}' \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \mid \langle \text{digit} \rangle \langle \text{identifier}' \rangle \mid \varepsilon \\ \langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\ \langle \text{constant}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \mid \varepsilon \\ \langle \text{letter} \rangle &\rightarrow a \mid \dots \mid z \mid A \mid \dots \mid Z \mid _ \\ \langle \text{digit} \rangle &\rightarrow 0 \mid 1 \mid \dots \mid 9 \\ \langle \text{character} \rangle &\rightarrow \text{any symbol except } " \end{aligned}$$

4 Theoretical Verification of Derivations

Instruction: `print("This is an example!!!!");`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{print_statement} \rangle ;$
 $\langle \text{print_statement} \rangle \rightarrow \text{print} (\langle \text{value} \rangle)$
 $\langle \text{value} \rangle \rightarrow \langle \text{literal} \rangle$
 $\langle \text{literal} \rangle \rightarrow " \langle \text{content} \rangle "$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{T}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{h}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{i}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{s}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow _$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{i}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{s}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow _$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{a}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{n}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow _$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{e}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{x}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{a}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{m}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$

$$\begin{aligned}
\langle \text{character} \rangle &\rightarrow p \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow l \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow e \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \varepsilon \\
\langle \text{statement_list} \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `int b = 20;`

$$\begin{aligned}
\langle \text{program} \rangle &\rightarrow \langle \text{statement_list} \rangle \\
\langle \text{statement_list} \rangle &\rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle \\
\langle \text{statement} \rangle &\rightarrow \langle \text{declaration} \rangle ; \\
\langle \text{declaration} \rangle &\rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle \\
\langle \text{type} \rangle &\rightarrow \text{int} \\
\langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\
\langle \text{letter} \rangle &\rightarrow b \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \\
\langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \\
\langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 2 \\
\langle \text{constant}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 0 \\
\langle \text{constant}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `float c = (500 * 4) / 20 + 1 - 10;`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{declaration} \rangle ;$
 $\langle \text{declaration} \rangle \rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle$
 $\langle \text{type} \rangle \rightarrow \text{float}$
 $\langle \text{identifier} \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{c}$
 $\langle \text{identifier}' \rangle \rightarrow \varepsilon$
 $\langle \text{value} \rangle \rightarrow \langle \text{expression} \rangle$
 $\langle \text{expression} \rangle \rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle$
 $\langle \text{term} \rangle \rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow (\langle \text{expression} \rangle)$
 $\langle \text{expression} \rangle \rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle$
 $\langle \text{term} \rangle \rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow \langle \text{constant} \rangle$
 $\langle \text{constant} \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 5$
 $\langle \text{constant}' \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 0$
 $\langle \text{constant}' \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 0$
 $\langle \text{constant}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow * \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow \langle \text{constant} \rangle$
 $\langle \text{constant} \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 4$
 $\langle \text{constant}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow \varepsilon$
 $\langle \text{expression}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow / \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow \langle \text{constant} \rangle$
 $\langle \text{constant} \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 2$
 $\langle \text{constant}' \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 0$
 $\langle \text{constant}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow \varepsilon$
 $\langle \text{expression}' \rangle \rightarrow + \langle \text{term} \rangle \langle \text{expression}' \rangle$

$$\begin{aligned}
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \\
\langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 1 \\
\langle \text{constant}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow -\langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \\
\langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 1 \\
\langle \text{constant}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 0 \\
\langle \text{constant}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `print(c);`

$$\begin{aligned}
\langle \text{program} \rangle &\rightarrow \langle \text{statement_list} \rangle \\
\langle \text{statement_list} \rangle &\rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle \\
\langle \text{statement} \rangle &\rightarrow \langle \text{print_statement} \rangle ; \\
\langle \text{print_statement} \rangle &\rightarrow \text{print} (\langle \text{value} \rangle) \\
\langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \\
\langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{identifier} \rangle \\
\langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\
\langle \text{letter} \rangle &\rightarrow c \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `print("hello");`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{print_statement} \rangle ;$
 $\langle \text{print_statement} \rangle \rightarrow \text{print} (\langle \text{value} \rangle)$
 $\langle \text{value} \rangle \rightarrow \langle \text{literal} \rangle$
 $\langle \text{literal} \rangle \rightarrow " \langle \text{content} \rangle "$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{h}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{e}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{l}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{l}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{o}$
 $\langle \text{content} \rangle \rightarrow \varepsilon$

Instruction: `string answer_5 = c;`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{declaration} \rangle ;$
 $\langle \text{declaration} \rangle \rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle$
 $\langle \text{type} \rangle \rightarrow \text{string}$
 $\langle \text{identifier} \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{a}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{n}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{s}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{w}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{e}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{r}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{-}$

$$\begin{aligned}
\langle \text{identifier}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{identifier}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 5 \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \\
\langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{identifier} \rangle \\
\langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\
\langle \text{letter} \rangle &\rightarrow c \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$